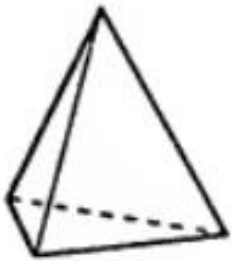


Eulerian and Hamiltonian Graphs

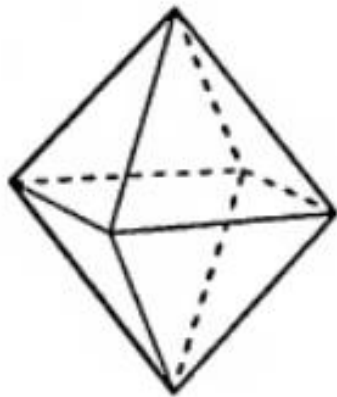
Mohammed Brahim



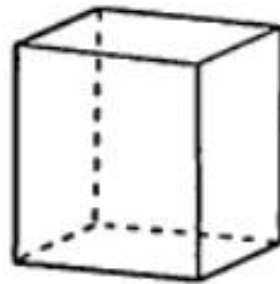
Platonic Graphs



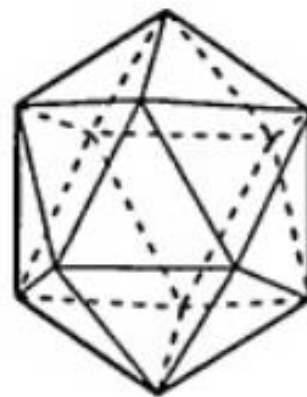
tetrahedron



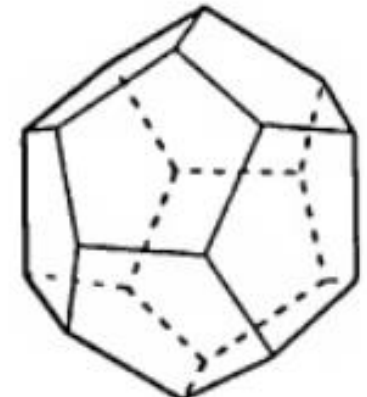
octahedron



cube



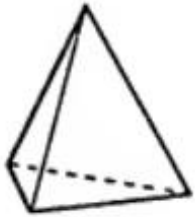
icosahedron



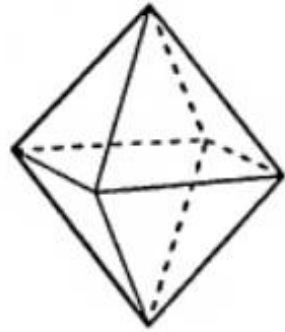
dodecahedron

How to start at one vertex and visit all edges before returning to it ?

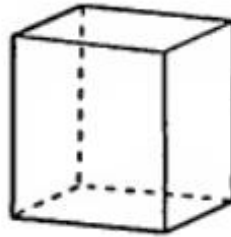
Platonic Graphs



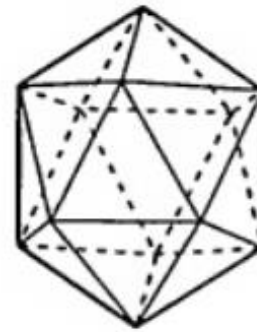
tetrahedron



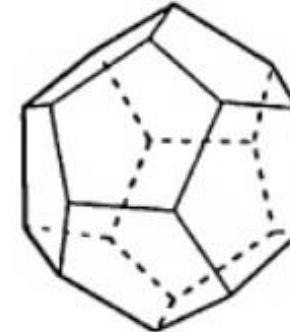
octahedron



cube



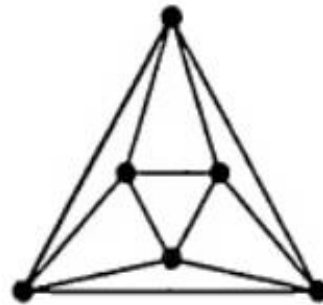
icosahedron



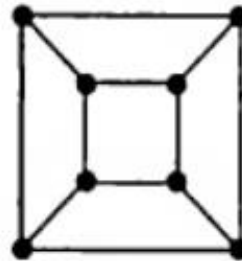
dodecahedron



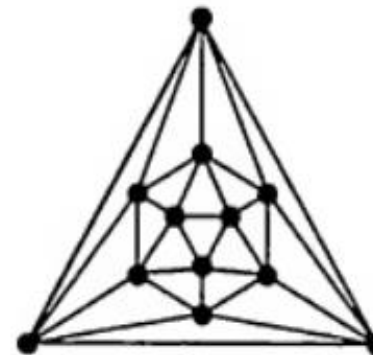
tetrahedron



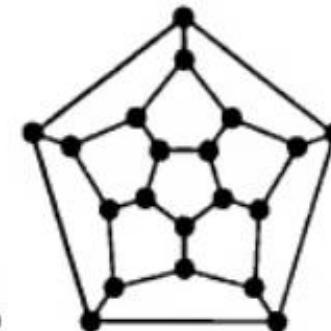
octahedron



cube

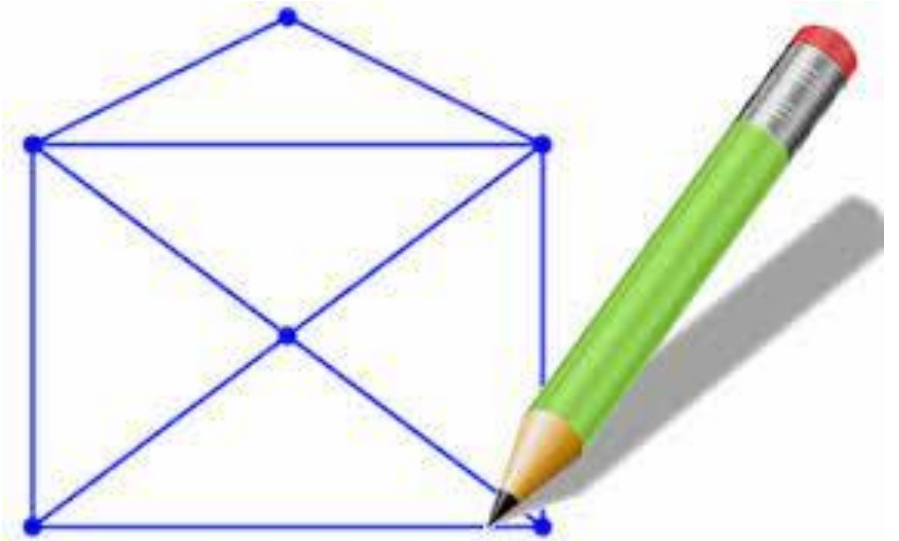
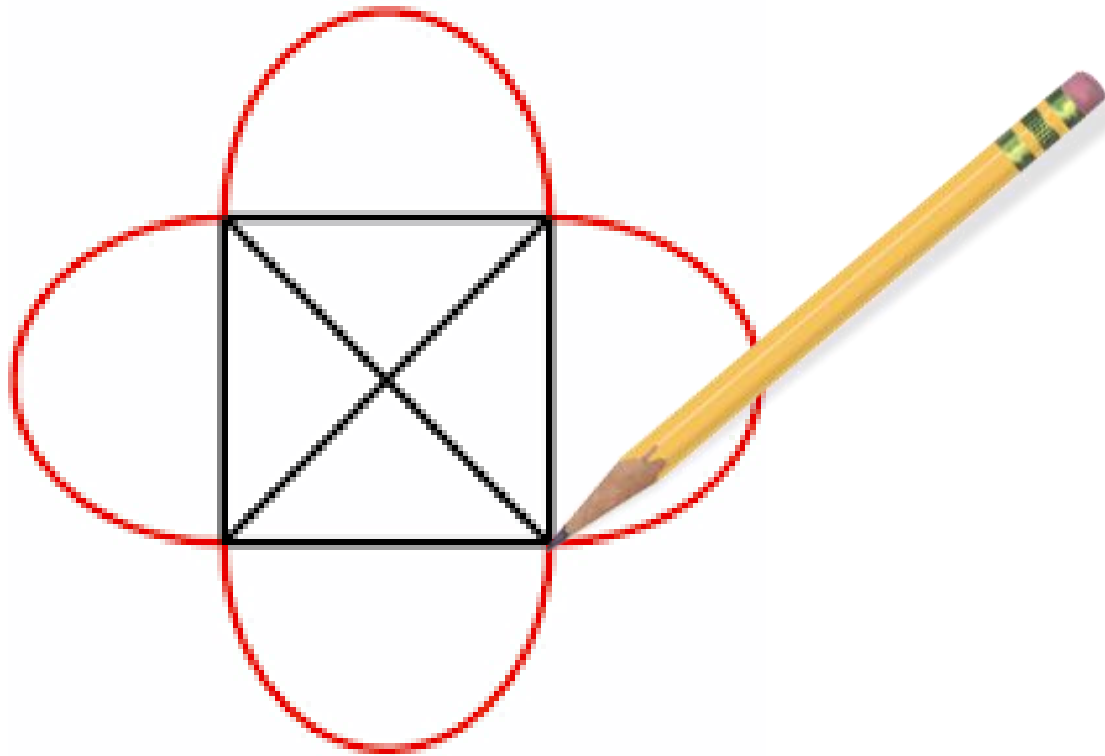


icosahedron

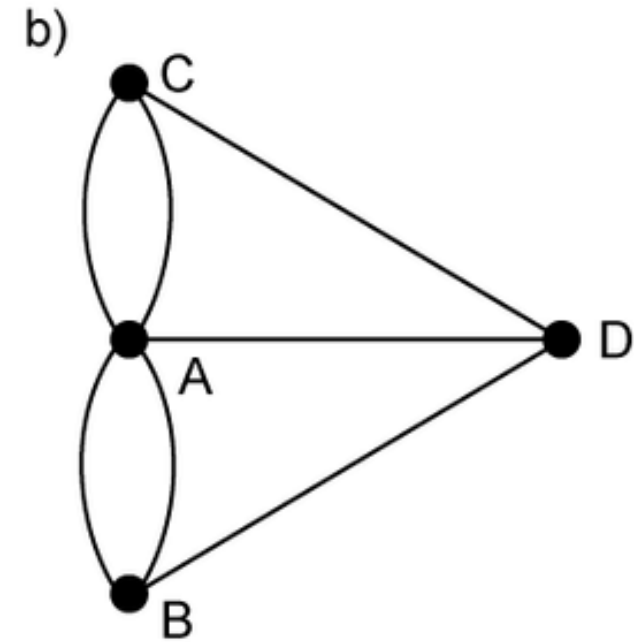
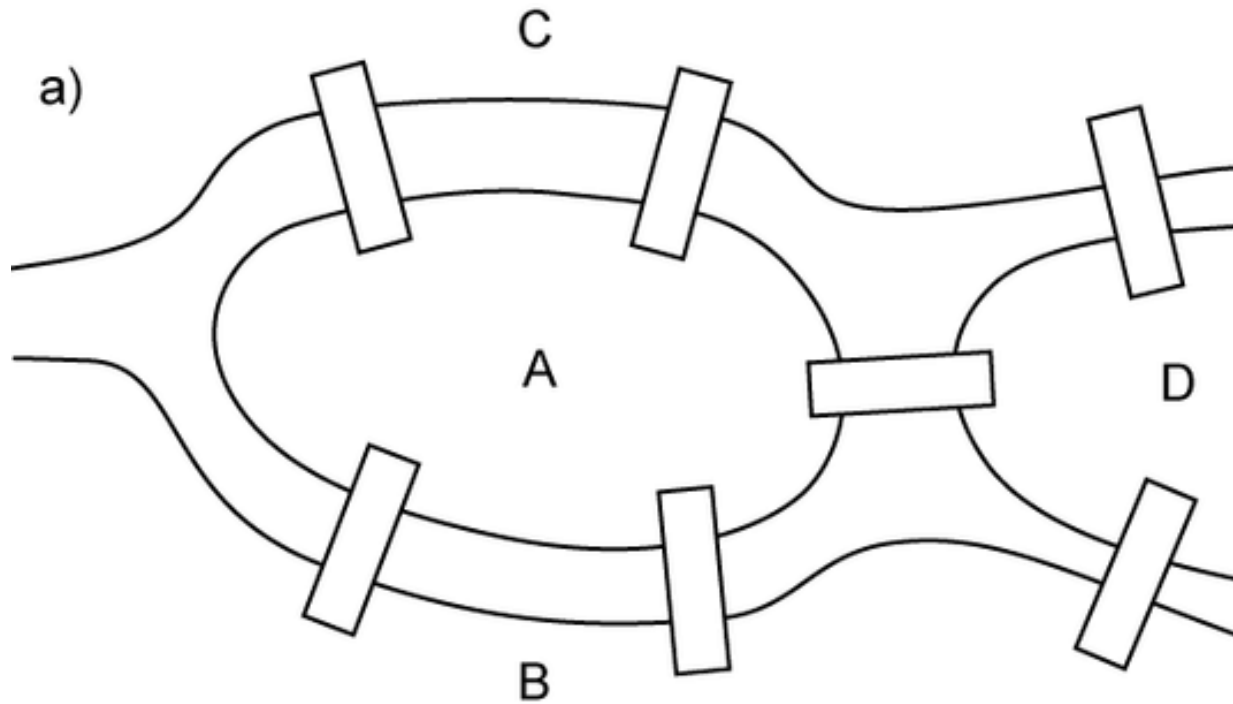


dodecahedron

Draw without lifting pencil puzzles



Konigsberg bridge problem



Explorer and Traveller

- **Explorer's Problem**

- An explorer wishes to find a tour that traverses each road exactly once and returns to the starting point.

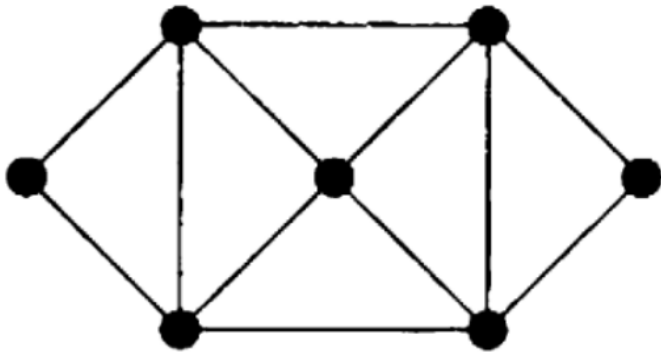
- **Traveller's Problem**

- A traveller wishes to find a tour that visits each city exactly once and returns to the starting point.

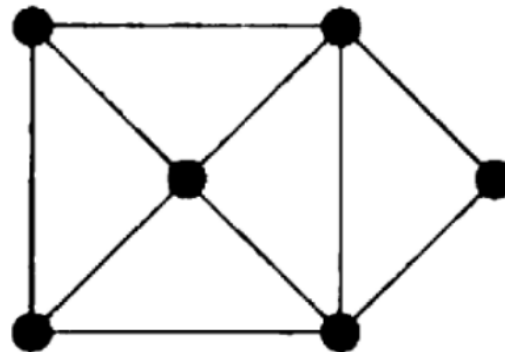
Eulerian vs Hamiltonian

- A connected graph is **Eulerian** if it contains a **closed trail that includes every edge**; such a trail is an **Eulerian trail**.
- A connected graph is **Semi-Eulerian** if there exists a (non-closed) trail that includes every edge of G .
- A connected graph is **Hamiltonian** if it contains a **cycle that includes every vertex**; such a cycle is a **Hamiltonian cycle**.
- A connected graph is **Semi-Hamiltonian** if there exists a (non-closed) **cycle that includes every vertex of G** .

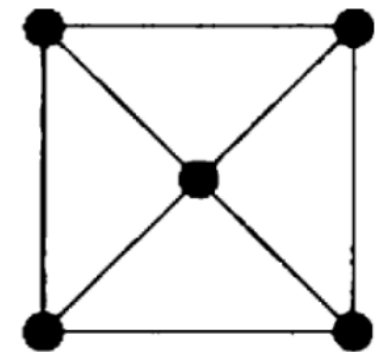
Eulerian ? Semi- Eulerian ?



Eulerian

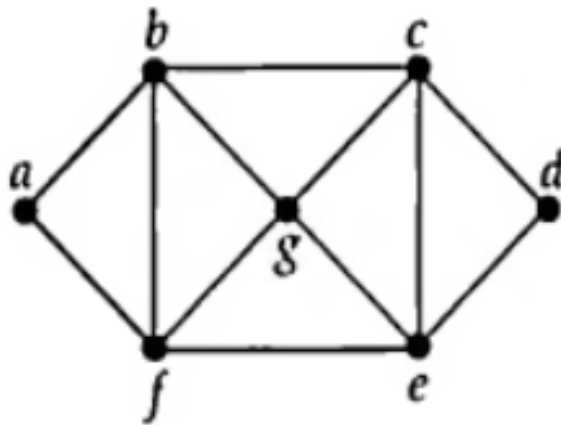


Semi-Eulerian



Non-Eulerian

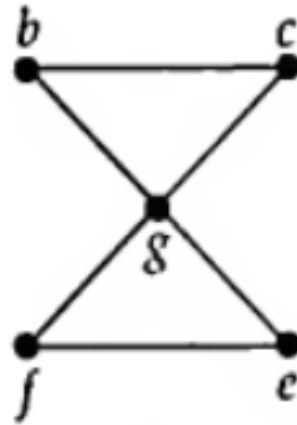
Hamiltonian ? Semi-Hamiltonian ?



(a)

Eulerian
Hamiltonian

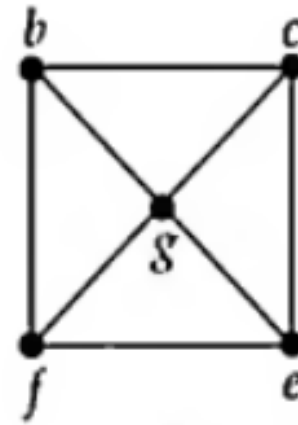
a b c d e f b g c e g f
a b c d e g f a



(b)

Eulerian
Semi-Hamiltonian

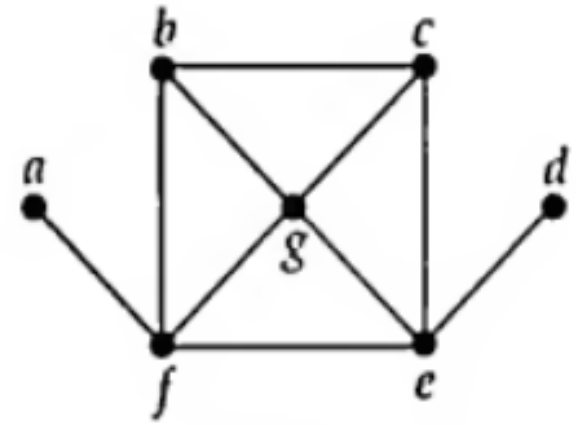
b c g f e g b



(c)

Not Eulerian
Hamiltonian

b c g e f b



(d)

Not Eulerian
Semi Hamiltonian

Eulerian Graphs and digraphs

Lemma 1

If G is graph in which the degree of each vertex is at least two, then G contains a cycle.

Proof of lemma1

- If G has loops or multiple edges, the proof is trivial, so assume G is a simple graph.
- Choose any vertex v from G .
- Construct a walk $V \rightarrow V_1 \rightarrow V_2 \rightarrow \dots$
 - At each step, choose a vertex V_{i+1} adjacent to V_i but not V_{i-1} already visited
 - This is possible because $\deg(V_i) \geq 2$.
 - Since G has a finite number of vertices, a vertex V_k will be repeated in the walk.
 - The part of the walk between the two occurrences of V_k forms a cycle.

Theorem 2 (Euler, 1735)

A connected graph is Eulerian if and only if each vertex has even degree.

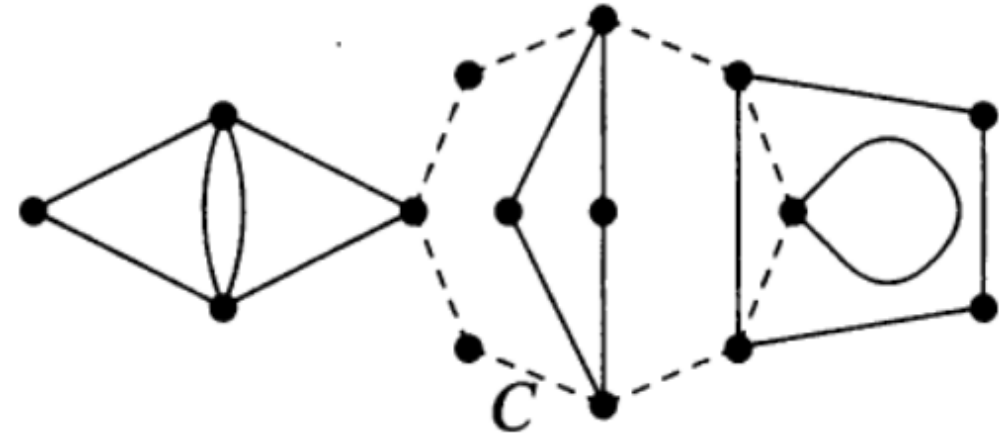
- This Theorem gives a **necessary and sufficient condition** for a connected graph to be Eulerian.
- It is equivalent to the following two statements for a connected graph G :
 - if G is Eulerian, then each vertex of G has even degree.
 - if each vertex of G has even degree, then G is Eulerian.

Proof →

- if G is Eulerian, then each vertex of G has even degree.
 - Let G be a graph and P be an Eulerian trail of G .
 - Whenever P passes through a vertex, its degree increased by 2.
 - Since each edge in P occurs exactly once, the degree of each vertex is even

Proof ←

- Proof by induction on the number of edges in G .
- Assume that each vertex of G has an even degree.
- G should contain a cycle C by Lemma 1.
- If C includes all edges of G , the proof is done.
- Otherwise, remove edges of C from G to create a new graph H with fewer edges.
 - Each vertex in H still has even degree.
 - By induction, each component of H has an Eulerian trail.
 - Each component of H has at least one vertex in common with C .
- Combine the initial cycle C and the cycles of each component to get the required Eulerian trail.



Corollary 1

An Eulerian graph can be split into cycles, no two of which have an edge in common.

- Proof
 - Similar to the proof of theorem 2 (direction ←)
 - Build initial cycle C
 - Remove the edges of this cycle
 - Build a cycle for each component

Corollary 2

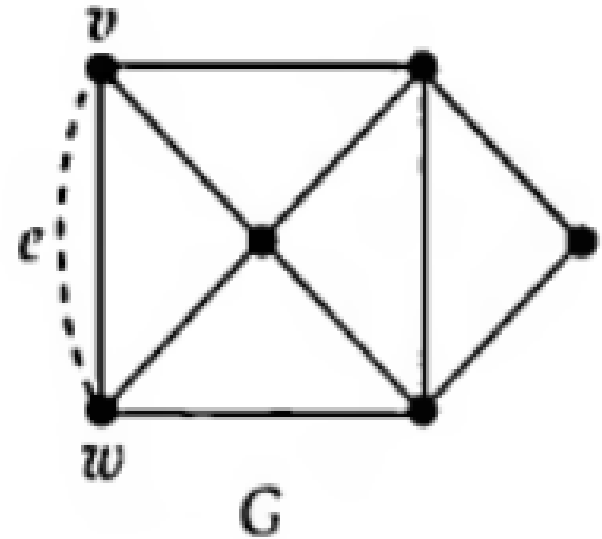
A connected graph is semi-Eulerian if and only if it has exactly two vertices of odd degree.

- A semi-Eulerian trail must start and end at vertices with odd degrees.
- According to the handshaking lemma, a graph cannot have only one vertex with an odd degree.

Proof ($\Rightarrow$)

If G is semi-Eulerian, then G has exactly two vertices of odd degree.

- Assume that G is semi-Eulerian graph.
- Add an edge e between the starting vertex v and finishing vertex w of an open trail.
- The result is an Eulerian graph in which all vertices have even degree (**Theorem 2**).
- By removing the edge e , we can return back to the semi-Eulerian graph G , **where only the vertices v and w have odd degrees.**



Proof (←)

*If a connected graph G has exactly two vertices of odd degree, then G is **semi-Eulerian**.*

- Let G be a graph with exactly two vertices of odd degree, v and w .
- Add an edge e joining v and w ; then we obtain a connected graph in which each vertex has even degree.
- This graph is Eulerian, and so has an Eulerian trail (**Theorem 2**).
- Removal of the edge e from this trail produces an open trail that includes every edge of G , so G is semi-Eulerian.

How to construct the Eulerian trail ?

Theorem 3 (Fleury's algorithm)

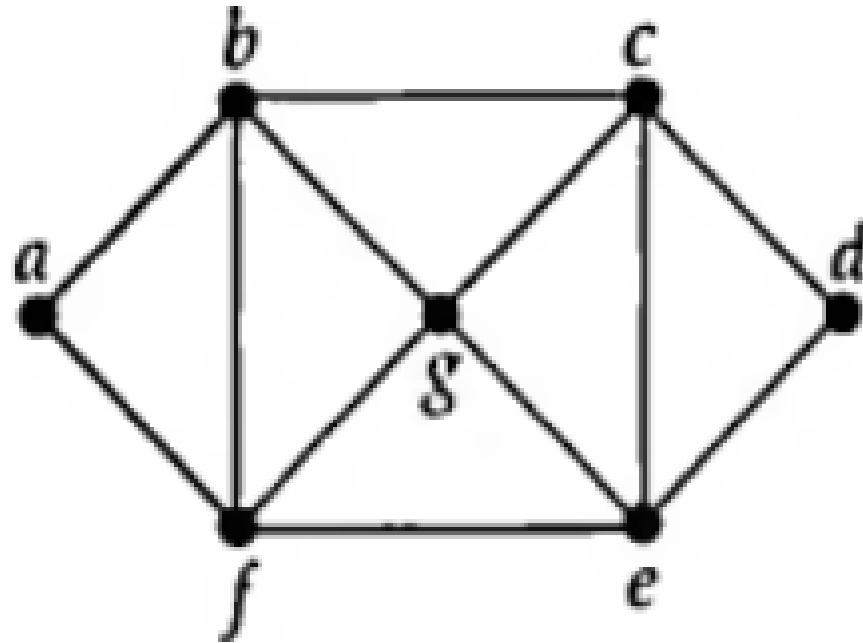
If G is an Eulerian graph. Then the following construction is always possible, and produce an Eulerian trail of G .

Start at any vertex and traverse the edges in an arbitrary manner, subject only to the following rules:

- 1. Erase the edges as they are traversed, and if any isolated vertices result, erase them too;*
- 2. At each stage, use a bridge only if there is no alternative.*

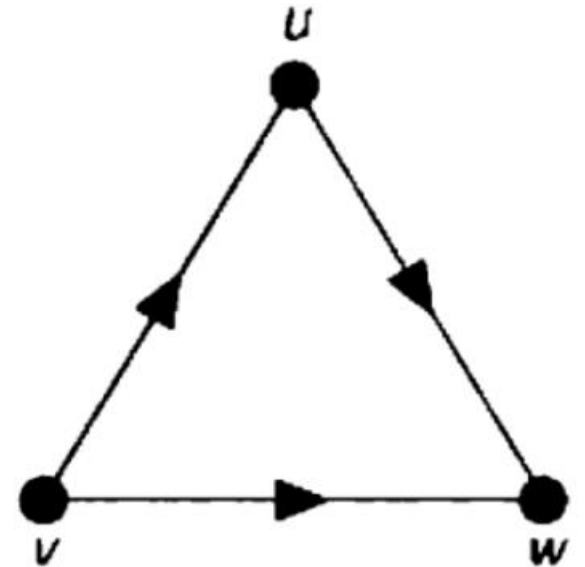
- Proof
 - Exercise

Example



Eulerian digraphs

- A connected digraph D is **Eulerian** if there exists a **closed directed trail that includes every arc of D** such a trail is an Eulerian trail.
- The digraph must be **strongly connected** for an Eulerian trail to exist.
- Any Eulerian graph is orientable, since we simply follow any Eulerian trail.



Theorem 4

A strongly connected digraph is Eulerian if and only if, for each vertex v of D

$$\mathbf{Outdeg}(v) = \mathbf{indeg}(v)$$

Proof

- Because the demonstration is completely similar to that of Theorem 2, it is left as an exercise.
- Defining Semi-Eulerian digraph and deriving digraph versions of Corollaries 1 and 2 is a simple.

Hamiltonian graphs and digraphs

Hamiltonian graphs

- We found the necessary and sufficient conditions for a connected graph to be Eulerian and aim to obtain similar characterizations for Hamiltonian graphs
- But **no such characterization is known**. In fact, little is known in general about Hamiltonian graphs, and most results have the form '**if G has enough edges, then G is Hamiltonian**'.
- Probably the most celebrated of these is Dirac's theorem, due to G. A. Dirac Corollary deduced it from the Ore Theorem.

Theorem 3 (Ore, 1960)

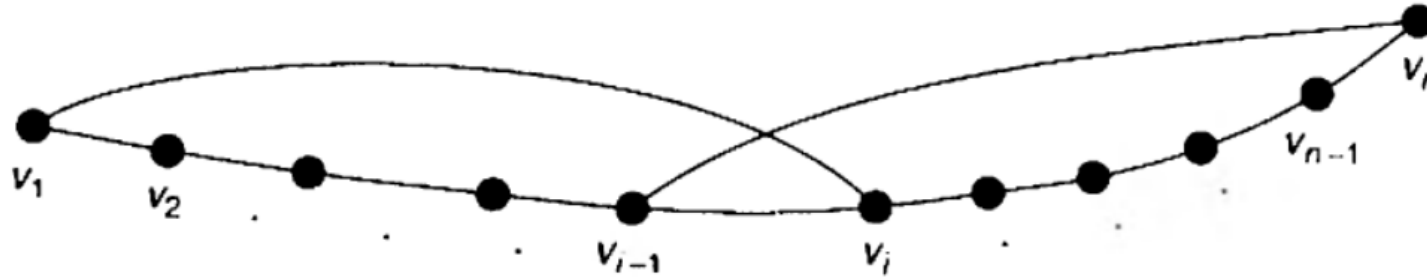
If G is simple graph with n (≥ 3) vertices and if

$$\deg(v) + \deg(w) \geq n$$

For each pair of non-adjacent vertices v and w , then G is Hamiltonian.

Proof

- Let G be a non-Hamiltonian graph with n vertices and the degree condition satisfied. Add extra edges if needed to make G "only just" non-Hamiltonian.
- G have a path $v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_n$ passing through every vertex. But v_1 and v_n are not adjacent because G is non-Hamiltonian.
- Because $\deg(v) + \deg(w) \geq n$
 - It exists some vertex v_i adjacent to v_1 with the property that v_{i-1} is adjacent to v_n .



- We can construct a Hamiltonian cycle:
$$v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_{i-1} \rightarrow v_n \rightarrow v_{n-1} \rightarrow \dots \rightarrow v_i \rightarrow v_1$$
- G' is Hamiltonian, contradicting the assumption that G is non-Hamiltonian.

Corollary 3 (Dirac, 1952)

If G is a simple graph with $n \geq 3$ vertices, and if $\deg(v) \geq \frac{1}{2}n$ for each vertex v , then G is Hamiltonian.

Proof:

- $\deg(v) + \deg(w) \geq \frac{1}{2}n + \frac{1}{2}n = n$
- Use Ore's theorem

Hamiltonian Digraph (M. A. Ghouila-Houri)

*Let D be a strongly connected digraph with n vertices if
 $\text{outdeg}(v) \geq \frac{1}{2}n$ and $\text{indeg}(v) \geq \frac{1}{2}n$ for each vertex v , then D is
Hamiltonian.*

Proof:

- Can be found **Douglas B. West, Introduction to Graph.**
- More difficult than Dirac corollary

Reference text book

